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WATER SUPPLY APPARATUS AND REFRIGERATOR COMPRISING SAME

Abstract

A refrigerator according to the present disclosure comprises: an ice-making apparatus that is disposed in a freezer chamber and produces ice; a water tank that is disposed in a refrigerating chamber formed above the freezer chamber and stores water to be supplied to the ice-making apparatus; a water supply pipe that connects the water tank and the ice-making apparatus to supply water from the water tank to the ice-making apparatus; a water supply pump provided in the water supply pipe to move the water in the water tank to the ice-making apparatus; and a bypass pipe of which one side diverges from the water supply pipe to the opposite direction of gravity and the other side is connected to the water tank so that water flowing back from the water supply pipe moves to the water tank.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Ser. No. 17/285,024 filed on Apr. 13, 2021, which is a U.S. National Stage Application which claims the benefit under 35 U.S.C. § 371 of International Patent Application No. PCT/KR2019/017811 filed on Dec. 16, 2019, which claims foreign priority benefit under 35 U.S.C. § 119 of Korean Patent Application 10-2018-0171150 filed on Dec. 27, 2018, in the Korean Intellectual Property Office. The contents of U.S. Ser. No. 17/285,024, International Patent Application No. PCT/KR2019/017811, and Korean Patent Application 10-2018-0171150 are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a water supply apparatus and a refrigerator comprising the same, and more particularly, to a water supply apparatus disposed in a refrigerating chamber and supplying water to an ice-making apparatus, and a refrigerator comprising the same.

BACKGROUND ART

[0003] In general, a refrigerator is a device that stores food fresh by including a storage space storing food and a cold air supply apparatus producing cold air through a refrigeration cycle and supplying the cold air to the storage space.

[0004] Based on a user's request, the refrigerator may include an ice-making apparatus producing ice. In order to operate such an ice-making apparatus, the refrigerator is required to include a water supply apparatus supplying water to the ice-making apparatus.

[0005] In general, in case of a French door refrigerator (FDR) type refrigerator having a refrigerating chamber on its upper side and a freezer chamber on its lower side, a water tank of the water supply apparatus may be installed in the refrigerating chamber and the ice-making apparatus disposed below the water tank may be installed in a freezer chamber. The water tank and the ice-making apparatus may be connected to each other by a water supply pipe including a water supply pump.

[0006] Water remaining in the water supply pipe may be frozen due to cold air introduced from the freezer chamber. In case that the water supply pipe is frozen, water in the water supply pipe may flow back to cause water leakage in the water supply apparatus.

DISCLOSURE

Technical Problem

[0007] The disclosure provides prevention of water leakage occurring in a water supply apparatus by allowing backflow water to be retrieved to a water tank in case that a backflow water phenomenon occurs in the water supply pipe.

Technical Solution

[0008] According to an embodiment in the disclosure, a refrigerator includes: an ice-making apparatus disposed in a freezer chamber and producing ice; a water tank disposed in a refrigerating chamber formed above the freezer chamber and storing water to be supplied to the ice-making

apparatus; a water supply pipe connecting the water tank and the ice-making apparatus to each other to supply water in the water tank to the ice-making apparatus; a water supply pump disposed on the water supply pipe and moving water in the water tank to the ice-making apparatus; and a bypass pipe having one side diverging from the water supply pipe in the opposite direction of gravity, and the other side connected to the water tank for water flowing back in the water supply pipe to move to the water tank.

[0009] The one side of the bypass pipe may be formed to have a predetermined height with respect to the water supply pipe.

[0010] The bypass pipe may be formed to be inclined downward from the one side to the other side.

[0011] The other side of the bypass pipe may be disposed on an upper portion of the water tank for the inside and outside of the water supply pipe to communicate with each other.

[0012] The bypass pipe may be disposed above a maximum level of water stored in the water tank to supply air to the water supply pipe.

[0013] The bypass pipe may be disposed at the rear end of the water supply pump to prevent water flowing back in the water supply pipe from moving to the water supply pump.

[0014] The bypass pipe may be formed to have a diameter smaller than a diameter of the water supply pipe.

[0015] The bypass pipe includes a T-type coupler disposed in the water supply pipe, and a connection line connecting the T-type coupler and the water tank to each other.

[0016] The T-type coupler may include a first connection portion formed along the water supply pipe, and a second connection portion extending from the first connection portion in the opposite direction of the gravity.

[0017] The second connection portion may be formed to have a predetermined height.

[0018] The connection line may be formed to be inclined downward from the second connection portion toward the water tank.

[0019] The water supply pipe may be disposed to be inclined downward toward the ice-making apparatus.

Advantageous Effects

[0020] The water supply apparatus having the above structure according to an embodiment of the disclosure may include the bypass pipe diverging from the water supply pipe disposed between the water tank and the ice-making apparatus, thereby allowing the backflow water to be retrieved to the water tank when the water is blocked from flowing in the water supply pipe, and also preventing water from overflowing to the outside of the water supply apparatus.

Description

DESCRIPTION OF DRAWINGS

[0021] FIG. 1 is a schematic front view of a refrigerator including a water supply apparatus according to an embodiment of the disclosure;

[0022] FIG. 2 is a partial perspective view of a refrigerator that shows the water supply apparatus and an ice-making apparatus according to an embodiment of the disclosure;

[0023] FIG. 3 is a perspective view of the water supply apparatus according to an embodiment of the disclosure;

[0024] FIG. 4 is a side view of the water supply apparatus according to an embodiment of the disclosure;

[0025] FIG. 5 is a cross-sectional view taken along “V-V” shown in FIG. 3; and

[0026] FIG. 6 is a cross-sectional view taken along “VI-VI” shown in FIG. 3.

BEST MODE

[0027] Hereinafter, embodiments of a water supply apparatus according to the disclosure and a refrigerator comprising the same are described in detail with reference to the accompanying drawings.

[0028] Embodiments described below are illustratively provided to assist in understanding of the disclosure, and it is to be understood that the disclosure may be variously modified and executed unlike the embodiments described herein. However, when it is decided that a detailed description for the known functions or components related to the disclosure may obscure the gist of the disclosure, the detailed description and concrete illustration will be omitted. Further, the accompanying drawings are not illustrated to scale, but sizes of some of components may be exaggerated to assist in the understanding of the disclosure.

[0029] Terms used in the specification, 'first', 'second', etc., may be used to describe various components, but the components are not to be interpreted to be limited to the terms. These terms may be used to differentiate one component from other components. For example, a 'first' component may be named a 'second' component and the 'second' component may also be similarly named the 'first' component, without departing from the scope of the disclosure.

[0030] Terms used in exemplary embodiments of the disclosure may be interpreted as the same meanings as meanings that are generally known to those skilled in the art unless defined otherwise.

[0031] In addition, terms such as 'fore end', 'rear end', 'upper portion', 'lower portion', 'upper end', 'lower end' and the like used in the disclosure are defined based on the drawings. The shapes and positions of respective components are not limited to these terms.

[0032] FIG. 1 is a schematic front view of a refrigerator including a water supply apparatus according to an embodiment of the disclosure.

[0033] The refrigerator 1 includes a main body 10, storage spaces 20 and 30 respectively disposed in the main body 10, doors 21 and 31 respectively opening and closing the storage spaces 20 and 30, an ice-making apparatus 200 disposed in the freezer chamber 30 and producing ice, a water supply apparatus 100 disposed in a refrigerating chamber 20 and supplying water to the ice-making apparatus 200, and the refrigeration cycle that providing cold air.

[0034] Here, the description describes that the ice-making apparatus 200 is disposed in the freezer chamber 30. However, the refrigerator 1 is not limited thereto, and may include an ice making room (not shown) in which the ice-making apparatus 200 is disposed. The ice-making apparatus 200 installed in the ice making room may be disposed below the water supply apparatus 100. Such an ice making room may be disposed at a corner in the refrigerating chamber or may be disposed at the rear of the door of the refrigerating chamber.

[0035] Refrigerator 1 may have a T-shaped partition wall 50 formed in the main body 10 and including horizontal and vertical partition walls. The storage spaces 20 and 30 may be partitioned into upper and lower portions by a horizontal partition wall, and the refrigerating chamber 20 may thus be disposed at the upper portion of the main body 10, and the freezer chamber 30 may be disposed at the lower portion of the main body 1. In addition, the freezer chamber 30 may be partitioned into left and right spaces by the T-shaped partition wall 50.

[0036] At least one shelf may be disposed in the refrigerating chamber 20, and food may be placed thereon.

[0037] A pair of the doors 21 of the refrigerating chamber may open and close the open front of the refrigerating chamber 20. The pair of the doors 21 of the refrigerating chamber may be hinged to both sides of the main body 1, and may respectively be pivoted forward. Each door 21 of the refrigerating chamber may have a handle disposed on its front, and may open and close the door 21 of the refrigerating chamber 20.

[0038] A pair of the doors 31 of the freezer chamber may open and close the open front of the freezer chamber 30. The pair of the doors 31 of the freezer chamber may be hinged to both sides of the main body 1, and may respectively be pivoted forward. Each door 31 of the freezer chamber

may have a handle disposed on its front, and may open and close the door **31** of the freezer chamber.

[0039] Meanwhile, the water supply apparatus **100** supplying water to the ice-making apparatus **200** may be disposed at a lower side of the refrigerating chamber **20**.

[0040] The ice-making apparatus **200** producing and storing ice by receiving water from the water supply apparatus **100** may be disposed at an upper side of the freezer chamber **30**.

[0041] A water supply pipe **150** of the water supply apparatus **100** may be connected to the ice-making apparatus **200**, and may supply water from the water supply apparatus **100** to the ice-making apparatus **200**.

[0042] FIG. **1** describes the T-type refrigerator having the T-shaped partition wall. However, the disclosure is not limited to this type, and may be applied to various types of refrigerators such as a so-called French door refrigerator (FDR) type refrigerator, in which a refrigerating chamber opened and closed by a pair of doors is disposed at its upper portion and a drawer-type freezer chamber is disposed at its lower portion, a bottom mounted freezer (BMF) type refrigerator and a 4-door refrigerator.

[0043] FIG. **2** is a partial perspective view of a refrigerator that shows the water supply apparatus and the ice-making apparatus according to an embodiment of the disclosure; and FIG. **3** is a perspective view of the water supply apparatus according to an embodiment of the disclosure.

[0044] Referring to FIGS. **2** and **3**, the water supply apparatus **100** may be disposed in the refrigerating chamber **20**, and the ice-making apparatus **200** may be disposed in the freezer chamber **30** formed below the refrigerating chamber **20**.

[0045] The water supply apparatus **100** may include a water tank **110**, the water supply pipe **150** connecting the water tank **110** and the ice-making apparatus **200** to each other, and a water supply pump **130** and a bypass pipe **170** diverged from the water supply pump **130**, and may thus supply water to the ice-making apparatus **200** which is described below.

[0046] The water tank **110** may store water to be supplied to the ice-making apparatus **200**. The water tank **110** may be filled with a predetermined amount of water.

[0047] The water supply pipe **150** may be formed to have one end connected to the water tank **110** and the other end connected to the ice-making apparatus **200**. Water stored in the water tank **110** may be supplied to the ice-making apparatus **200** through the water supply pipe **150**.

[0048] One end **130a** of the water supply pipe **150** may be connected to the water tank **110**. The one end **130a** of the water supply pipe **150**, which transfers water from the water tank **110** to the ice-making apparatus **200**, may be disposed to be spaced apart from one end **170a** of the bypass pipe **170** which is described below.

[0049] The water supply pump **130** may be installed on the water supply pipe **150**. The water supply pump **130** may pump water stored in the water tank **110** and supply water to the ice-making apparatus **200**. The water supply pump **130** may be configured of a pump which may be rotated in one direction. In case that the water supply pump **130** is rotated, water stored in the water tank **110** may be supplied to the ice-making apparatus **200** along the water supply pipe **150** by suction power of the water supply pump **130**.

[0050] That is, water stored in the water tank **110** may be supplied to the ice-making apparatus **200** by passing through the water supply pump **130** and the water supply pipe **150** in sequence.

[0051] If the water supply pump **130** is operated and water from the water tank **110** moves to the ice-making apparatus **200** through the water supply pipe **150**, water provided to the ice-making apparatus **200** may be frozen in the ice-making apparatus **200** to be converted to ice.

[0052] Then, if the water supply pump **130** stops its operation, water in the water tank **110** may no longer move to the ice-making apparatus **200**. Here, water that has yet to flow out of the pipe may remain at both the ends of the water supply pipe **150**, and a vacuum pressure may be generated between the ends.

[0053] The bypass pipe **170** may guide water flowing back in the water supply pipe **150** to move to

the water tank **110**. The bypass pipe **170** may have one side diverging from the water supply pipe **150** in the opposite direction of gravity, and the other side connected to the water tank **110**.

[0054] In case that water has yet to flow out of the water supply pipe **150** and remains therein, the water supply pipe **150** may be frozen by cold air introduced from the freezer chamber **30**. In case that the water supply pipe **150** is frozen, water flowing back in the water supply pipe **150** may be retrieved into the water tank **110** along the bypass pipe **170**.

[0055] In case that the water supply pipe **150** is frozen, water in the water supply pipe **150** may flow back, and water may thus leak from a relatively weak portion of the water supply pipe **150**, around the water supply pump **130**. The bypass pipe **170** may guide the water flowing back in the water supply pipe **150** to move back to the water tank **110**, thereby preventing the water leakage in the water supply apparatus **100**.

[0056] In addition, the bypass pipe **170** may be disposed to be exposed to the outside for the inside and outside of the water supply pipe **150** to communicate with each other. That is, the bypass pipe **170** may be formed to have the other side open to external air for the air to be introduced into the water supply pipe **150**.

[0057] In detail, the other side **170a** of the bypass pipe **170** may be formed to be connected to an upper portion of the water tank **110**. The other side **170a** of the bypass pipe **170** may be disposed above a maximum level of water stored in the water tank **110**.

[0058] If water is supplied to the ice-making apparatus **200** by the water supply pump **130** and then the water supply pump **130** stops its operation, water may no longer move from the water tank **110** to the ice-making apparatus **200**. Here, water that has yet to flow out of the pipe may remain at both the ends of the water supply pipe **150**, and the vacuum pressure may be generated between the ends. As the vacuum pressure is generated in the water supply pipe **150**, water in the water tank **110** cannot continue to move to the ice-making apparatus **200**.

[0059] In this case, as the present disclosure has the bypass pipe **170** diverging from the water supply pipe **150**, it is possible to prevent such a vacuum pressure from being generated between both the ends of the water supply pipe **150** due to air injected from the bypass pipe **170**.

Accordingly, water head difference in the water supply pipe **150** may be maintained based on a height difference between the water tank **110** and the ice-making apparatus **200**.

[0060] In addition, the bypass pipe **170** may provide air to the water supply pipe **150**, and may thus prevent water from remaining in the water supply pipe **150** in case that water supply is stopped.

[0061] The ice-making apparatus **200** receiving water from the water supply apparatus **100** may be disposed in a lower position than the water supply apparatus **100**. In detail, an ice-making tray **210** of the ice-making apparatus **200** receiving water may be disposed at the lower position than the water tank **110** of the water supply apparatus **100**.

[0062] The ice-making apparatus **200** may include the ice-making tray **210** and an ice storage **230** storing produced ice.

[0063] The ice-making tray **210** is a bowl in which ice is produced, and may be formed to have an open upper surface for the water supplied from the water tank **110** to be supplied thereto.

[0064] A water supply **151** of the water supply pipe **150** may be disposed at one side of the ice-making tray **210**, and may supply water to the ice-making tray **210**.

[0065] A refrigerant pipe may be disposed at a lower portion of the ice-making tray **210** while being in contact therewith. The ice-making tray **210** may include an ejector (not shown) pushing out ice produced from the ice-making tray **210** and discharging the ice from the ice-making tray **210**.

[0066] The ice storage **230** may have a shape of a box having an open upper surface to store ice discharged from the ice-making tray **210** by the ejector, and may be disposed below the ice-making tray **210**.

[0067] FIG. **4** is a side view of the water supply apparatus according to an embodiment of the disclosure.

[0068] Referring to FIG. 4, the water supply pipe **150** may include a first flow path **131** through which water discharged from the water tank **110** is introduced to the water supply pump **130** and a second flow path **133** through which water is discharged from the water supply pump **130**.

[0069] Water stored in the water tank **110** may move to the water supply pump **130** along the first flow path **131** by the suction power of the water supply pump **130**, and water passed through the water supply pump **130** may be supplied to the ice-making apparatus **200** along the second flow path **133**.

[0070] The second flow path **133** and the water supply pipe **150** may be connected to each other, and may be formed integrally with each other.

[0071] The bypass pipe **170** may be formed between the second flow path **133** and the water supply pipe **150**. The bypass pipe **170** may be disposed at the rear end of the water supply pump **130** to prevent water flowing back in the water supply pipe **150** from moving to the water supply pump **130**.

[0072] The bypass pipe **170** may be formed to diverge from the water supply pipe **150** in the opposite direction of the gravity.

[0073] The bypass pipe **170** may include a T-type coupler **171** disposed in the water supply pipe **150**, and a connection line **175** connecting the T-type coupler **171** and the water tank **110** to each other.

[0074] The T-type coupler **171** may include a first connection portion **172** formed along the water supply pipe **150** and a second connection portion **173** extending vertically from the first connection portion **172**.

[0075] The first connection portion **172** may be formed horizontally with respect to a portion of the water supply pipe **150**, and the second connection portion **173** may be formed vertically with respect to the first connection portion **172** and may have a predetermined height.

[0076] The second connection portion **173** may be formed perpendicular to a portion of the water supply pipe **150**, and may communicate with the first connection portion **172**. The second connection portion **173** may extend from the first connection portion **172** in the opposite direction of the gravity. A point where the second connection portion **173** and the first connection portion **172** meet with each other may correspond to a diverging point of the T-type coupler **171**.

[0077] The first connection portion **172** and the second connection portion **173** may be formed integrally with each other.

[0078] The second connection portion **173** may have one end communicating with the first connection portion **172**, and may have the other end at which the connection line **175** is disposed. Water overflowing from the second connection portion **173** having the predetermined height may be discharged to the water tank **110** along the connection line **175**.

[0079] The connection line **175** may be formed from the T-type coupler **171** to be inclined downward toward the water tank **110**. Water introduced to the connection line **175** may be discharged to the water tank **110** by the gravity.

[0080] The bypass pipe **170** may be formed to have the predetermined height with respect to the water supply pipe **150**. Accordingly, in case that the water supply pump **130** is operated, the water moving along the second flow path **133** may not move to the bypass pipe **170**, and may move to the ice-making apparatus **200** along the water supply pipe **150**.

[0081] Meanwhile, in a process in which a large amount of water moves by the water supply pump **130**, the water supply pipe **150** may be frozen, and the other side of the water supply pipe **150** (its region adjacent to the ice-making apparatus) may thus be blocked. In this case, water may fill the water supply pipe **150** and the second flow path **133**. Here, the water filling the water supply pipe **150** and the second flow path **133** may overflow to the second connection portion **173** of the bypass pipe **170** and may move along the bypass pipe **170**. That is, in case that the water supply pipe **150** is frozen and water flows back, the backflow water may move to the water tank **110** by the bypass pipe **170**.

[0082] In case that a backflow water phenomenon occurs in the water supply pipe **150**, the backflow water may move to the bypass pipe **170** diverged upward from the water supply pipe **150**. Accordingly, it is thus possible to prevent water from leaking from the water supply pipe **150** to the outside of the water supply apparatus **100**.

[0083] The water flowing back in the water supply pipe **150** may not move to the water supply pump **130**, and may be retrieved to the water tank **110** at the diverging point of the T-type coupler **171**.

[0084] FIG. **5** is a cross-sectional view taken along “V-V” shown in FIG. **3**.

[0085] Referring to FIG. **5**, the bypass pipe **170** may be disposed on the upper portion of the water tank **110**. The bypass pipe **170** may include an air opening **176** supplying air to the water supply pipe **150**.

[0086] The air opening **176** may be formed in the other side of the bypass pipe **170**.

[0087] The air opening **176** may be disposed above the maximum level of water stored in the water tank **110**. The air opening **176** may be disposed to be exposed to the outside to supply air to the water supply pipe **150**. That is, the air opening **176** may be disposed in the upper portion of the water tank **110** where water does not reach.

[0088] The air opening **176** may be disposed above the maximum level of water stored in the water tank **110**. Accordingly, water may not be supplied to the bypass pipe **170** if the water supply pump **130** is operated, and air may be introduced to the bypass pipe **170** if the water supply pump **130** stops its operation.

[0089] External air may be introduced into the water supply pipe **150** through the air opening **176**.

[0090] It is possible to prevent the vacuum pressure from being generated in the water supply pipe **150** and to prevent water from staying in the pipe by forming the bypass pipe **170** including the air opening **176** exposed to air.

[0091] The water supply pipe **150** may communicate with the bypass pipe **170** including the air opening **176**. An atmospheric pressure may be applied to the water supply pipe **150** through the bypass pipe **170** including the air opening **176**, and accordingly, water may not stay in the water supply pipe **150** due to the pressure caused by the water head difference in the water supply pipe **150**.

[0092] In addition, if water is supplied to the ice-making apparatus **200** by the water supply pump **130**, and then the water supply pump **130** stops its operation, water that has yet to flow out of the water supply pipe **150** may remain at both the ends of the pipe. Here, air may be introduced to the water supply pipe **150** through the air opening **176** of the bypass pipe **170** to discharge water remaining in the water supply pipe **150**.

[0093] As water remaining in the water supply pipe **150** is discharged, it is possible to solve the problem in which water remains in the pipe due to the water head difference in the water supply pipe **150**. It is also possible to drain water remaining in the water supply pipe **150**, and may thus reduce a possibility in which the water supply pipe **150** is frozen.

[0094] Meanwhile, among the sides of the water supply pipe **150**, the other side of the water supply pipe **150**, which is adjacent to the ice-making apparatus **200**, may be frozen, and water in the water supply pipe **150** may thus flow back. Even in this case, backflow water may be retrieved to the water tank **110** along the bypass pipe **170**.

[0095] In case that one side of the water supply pipe **150** is blocked by freezing or the like, water discharged from the water supply pump **130** may fill the T-type coupler **171**, and water filling the T-type coupler **171** may move to the connection line **175**. In detail, water may fill up to an upper portion of the first connection portion **172** and may fill up to the second connection portion **173** connected to the first connection portion **172**. Water overflowing from the second connection portion **173** may move to the water tank **110** along the connection line **175**.

[0096] The connection line **175** may be disposed to be inclined downward toward the water tank **110**. Water introduced to the connection line **175** may move to the water tank **110** by the gravity.

[0097] A guide portion **180** guiding water discharged from the connection line **175** to be discharged to the water tank **110** may be disposed at the other side of the connection line **175**.

[0098] The guide portion **180** may be formed to surround the air opening **176** of the connection line **175** to prevent the air opening **176** from being blocked by water or a foreign material. A portion of the connection line **175** may be disposed inside the guide portion **180**.

[0099] The guide portion **180** may be formed to be spaced apart from the air opening **176** by a predetermined distance.

[0100] The guide portion **180** may protrude downward to guide water discharged from the air opening **176** to move downward.

[0101] In case that the other side of the water supply pipe **150** is blocked by freezing or the like of the water supply pipe **150**, the bypass pipe **170** may guide water flowing back in the water supply pipe **150** to move back to the water tank **110**, thereby preventing water leakage from occurring in the water supply apparatus **100**.

[0102] FIG. **6** is a cross-sectional view taken along “VI-VI” shown in FIG. **3**.

[0103] Referring to FIG. **6**, the bypass pipe **170** may be formed to have a diameter smaller than a diameter of the water supply pipe **150**. The bypass pipe **170** may be formed to have the diameter smaller than the water supply pipe **150** to prevent water supplied to the ice-making apparatus **200** from spilling through the bypass pipe **170** in the process in which a large amount of water moves by the water supply pump **130**.

[0104] In detail, the second connection portion **173** of the bypass pipe **170**, which is formed to have the predetermined height, may have a diameter smaller than the water supply pipe **150**.

[0105] To produce ice, the water supply pump **130** may supply water from the water tank **110** to the ice-making apparatus **200**. If the water supply pump **130** stops its operation after a predetermined period of the operation, air may be introduced to the water supply pipe **150** through the bypass pipe **170**. The introduced air may discharge water remaining in the water supply pipe **150** to the ice-making apparatus **200**.

[0106] If all water remaining in the water supply pipe **150** is discharged to the ice-making apparatus **200**, it is possible to prevent water from remaining in the water supply pipe **150**.

[0107] In addition, the bypass pipe **170** may be disposed at a position above the maximum level of water stored in the water tank **110**. Accordingly, water may not move from the bypass pipe **170** to the water supply pipe **150** when the water supply pump **130** is operated to supply water to the ice-making apparatus **200**, and air may be introduced from the bypass pipe **170** to the water supply pipe **150** when the water supply pump **130** stops its operation. It is possible to discharge all water remaining in the water supply pipe **150** by the introduced air, thereby preventing water from remaining in the pipe to obstruct a water flow, in advance.

[0108] In addition, the bypass pipe **170** may diverge from the water supply pipe **150** in the opposite direction of the gravity. Accordingly, if the backflow water phenomenon occurs in the water supply pipe **150**, the backflow water may be retrieved to the water tank **110** along the diverging bypass pipe **170**, thereby preventing the water leakage from occurring in the water supply apparatus **100**.

[0109] Hereinabove, the disclosure is described as an illustrative method. It is to be understood that terms used herein are provided to describe the disclosure rather than limiting the disclosure. Various modifications and alternations of the disclosure may be made according to the contents described above. Therefore, the disclosure may be freely practiced without departing from the scope of the claims unless additionally mentioned.

Claims

1. A refrigerator comprising: an ice-making apparatus disposed in a freezer chamber and configured to produce ice; a water tank disposed in a refrigerating chamber formed above the freezer chamber and configured to store water to be supplied to the ice-making apparatus; a water supply pipe

connecting the water tank and the ice-making apparatus to each other to supply water in the water tank to the ice-making apparatus; a water supply pump disposed on the water supply pipe and configured to move water in the water tank to the ice-making apparatus; and a bypass pipe having one side diverging from the water supply pipe in an opposite direction of gravity, and an other side that longitudinally extends horizontally along an upper portion of the water tank above a maximum water level in the water tank and that has an end that is open for water flowing back in the water supply pipe to move therethrough to the water tank, wherein the water tank includes a guide portion disposed on an upper portion of the water tank, the guide portion has one end longitudinally extending horizontally and surrounding the other side of the bypass pipe, spaced apart from an outer circumference of the other side of the bypass pipe, and another end protruding downward from the one end, so that the guide portion surrounds the end of the other side of the bypass pipe that is open, and the bypass pipe is above the water supply pipe.

2. The refrigerator as claimed in claim 1, wherein the bypass pipe has the other side that has the end that is open for external air to be introduced into the water supply pipe.
 3. The refrigerator as claimed in claim 1, wherein the bypass pipe is inclined downward from the one side to the other side.
 4. The refrigerator as claimed in claim 1, wherein the bypass pipe includes a T-type coupler disposed in the water supply pipe and includes a first connection portion formed along the water supply pipe and a second connection portion extending from the first connection portion in the opposite direction of the gravity.
 5. The refrigerator as claimed in claim 4, wherein the second connection portion of the bypass pipe has a diameter smaller than the water supply pipe.
 6. The refrigerator as claimed in claim 1, wherein the bypass pipe is disposed at a rear end of the water supply pump to prevent water flowing back in the water supply pipe from moving to the water supply pump.
 7. The refrigerator as claimed in claim 1, wherein the bypass pipe has a diameter smaller than a diameter of the water supply pipe.
 8. The refrigerator as claimed in claim 4, wherein the bypass pipe includes a connection line connecting the T-type coupler and the water tank to each other.
 9. The refrigerator as claimed in claim 8, wherein the connection line is inclined downward from the second connection portion toward the water tank.
 10. The refrigerator as claimed in claim 1, wherein the water supply pipe is inclined downward toward the ice-making apparatus.
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